Chapter

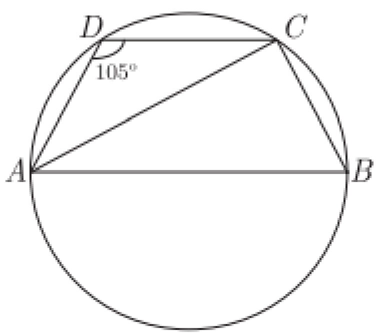
Cyclic Properties Of Circle

---

1. ABCD is a cyclic quadrilateral. Diameter AB is parallel to DC and +ADC = 105º. SQP 2023

(i) Find +ABC .

(ii) Find +ACD.

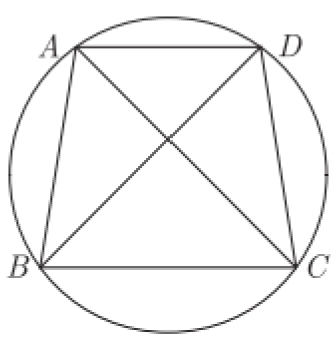


2. In the given figure, ABCD is a cyclic quadrilateral. +ABC = 70°, +ACB = 50° and +CAD = 30°.

(i) Find +BAC .

(i) Find +ADC .

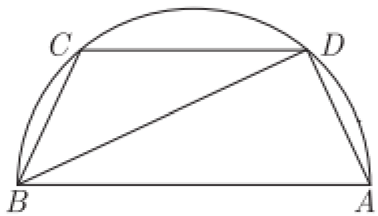
(iii) Prove that BD is a diameter. Comp 2023



3. In given figure C and D are points on the semi-circle described on BA as diameter. Given +BAD = 70° and +DBC = 30º.

(i) Find +ABD.

(ii) Find +BDC . Main 2022


C
D
B
A

Solution

---

1. Given:

- ABCD is a cyclic quadrilateral
- AB is a **diameter**
- $AB \parallel DC$
- $\angle ADC = 105^\circ$

(i) Find $\angle ABC$

In a cyclic quadrilateral, opposite angles are supplementary:

$$\angle ABC + \angle ADC = 180^\circ$$

$$\angle ABC = 180^\circ - 105^\circ = \boxed{75^\circ}$$

(ii) Find $\angle ACD$

Since **AB is a diameter**,

$\angle ACB = 90^\circ$

Also, because $AB \parallel DC$, the angle between AC and DC equals the angle between AC and AB:

$\angle ACD = \angle CAB$

Now in $\triangle ABC$:

$\angle CAB = 180^\circ - (90^\circ + 75^\circ) = 15^\circ$

So,

$\angle ACD = 15^\circ$

Final Answers:

- (i) $\angle ABC = 75^\circ$
- (ii) $\angle ACD = 15^\circ$

2.

Given: $ABCD$ cyclic, $\angle ABC = 70^\circ,\ \angle ACB = 50^\circ,\ \angle CAD = 30^\circ$.

(i) Find $\angle BAC$

In $\triangle ABC$:

$$\angle BAC = 180^\circ - (70^\circ + 50^\circ) = \boxed{60^\circ}$$

(ii) Find $\angle ADC$

Opposite angles of a cyclic quadrilateral are supplementary:

$$\angle ADC = 180^\circ - \angle ABC = 180^\circ - 70^\circ = \boxed{110^\circ}$$

(iii) Prove BD is a diameter

At A, AC lies inside $\angle BAD$, so

$$\angle BAD = \angle BAC + \angle CAD = 60^\circ + 30^\circ = \boxed{90^\circ}$$

An angle subtending chord BD is a right angle $\Rightarrow$ BD is a **diameter**. ✅

3. (i) BA is a diameter $\Longrightarrow \angle BDA = 90^\circ$

In $\triangle BAD$:

$\angle ABD = 180^\circ - (90^\circ + 70^\circ) = 20^\circ$

(ii) $\angle BDC$ and $\angle BAC$ subtend the same arc BC.

$\angle BDC = 2 \times \angle ABD = 2 \times 20^\circ = 40^\circ$

Answers:

(i) 20°

(ii) 40°